

JOURNAL CLUB

# NATURE Trial

Mechanical thrombectomy with enVast in STEMI with large thrombus burden — a randomised superiority trial (ESC Congress 2026)

Presented by

**Dr M. Waqas Saleem**

PGR Cardiology

Cardiac Centre, BVH Bahawalpur



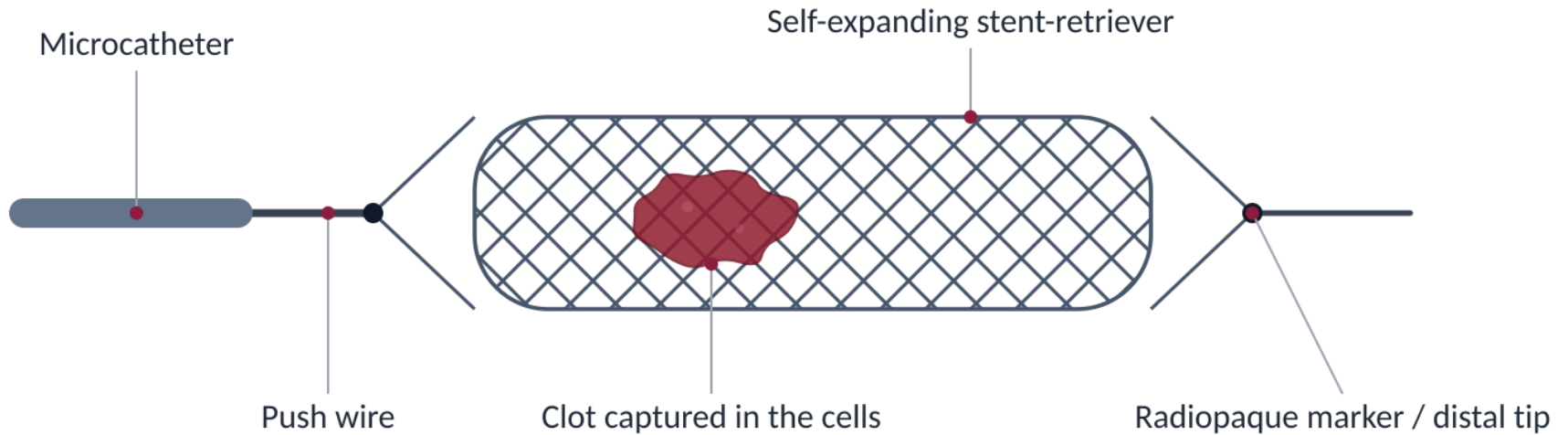
# Background: why a large thrombus matters



## What we already know

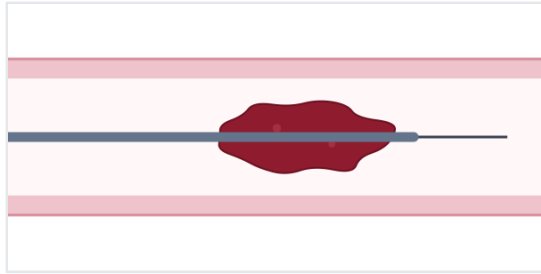
- TASTE (n = 7,244): routine aspiration did not reduce mortality
- TOTAL (n = 10,732): no clinical benefit, and more strokes at 30 days
- ESC guidelines: routine thrombus aspiration is not recommended (Class III)
- Question: can a stent-retriever remove clot better and more safely?

# The enVast device



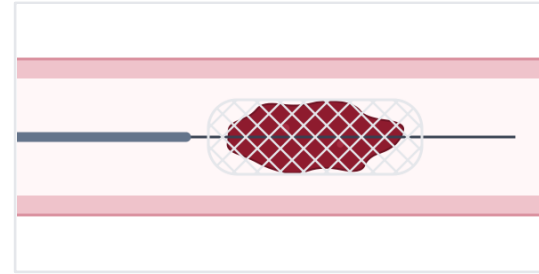
*Schematic illustration (not to scale). Coronary stent-retriever by Vesalio, derived from the NeVa neurovascular thrombectomy platform.*

# The procedure step by step



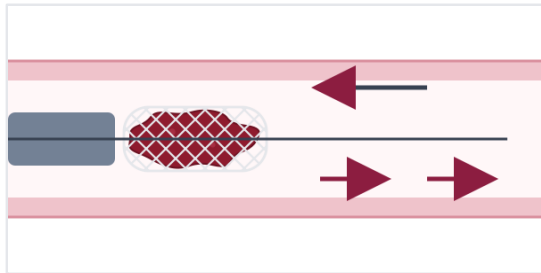
## 1 Cross

Wire and microcatheter pass through the clot



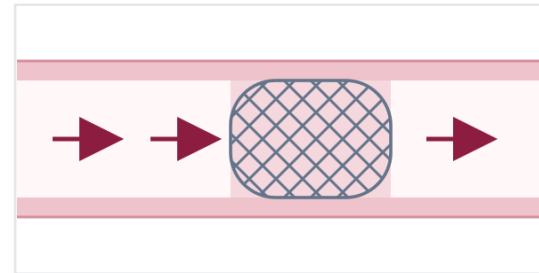
## 2 Deploy

Retriever opens inside the clot and holds it



## 3 Retrieve

Device and clot pulled back into the guide

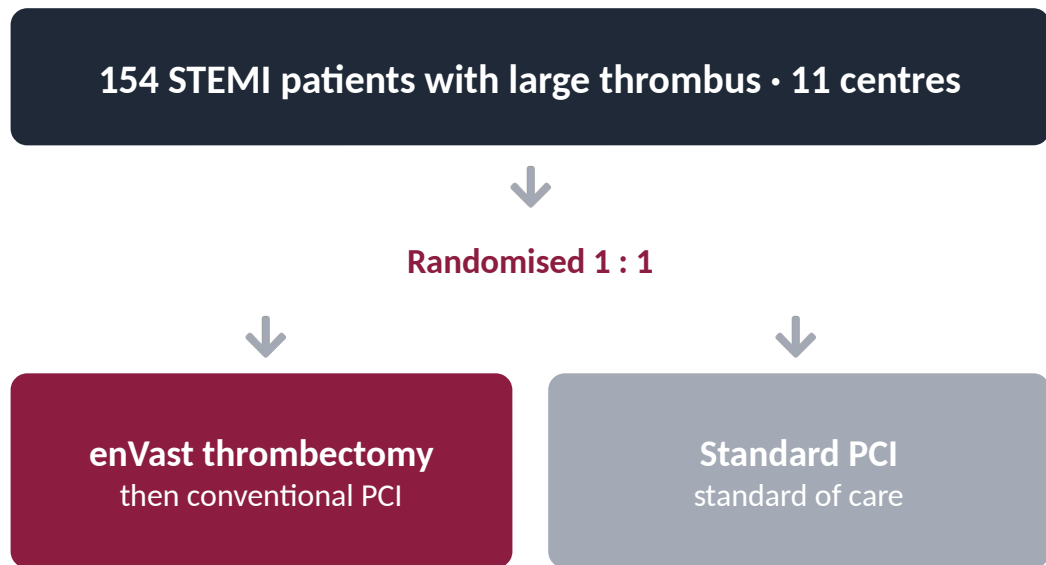


## 4 Stent

Conventional PCI; flow restored

*Schematic illustrations (not to scale).*

# Trial design



|            |                                      |
|------------|--------------------------------------|
| Type       | Prospective, multicentre, randomised |
| Hypothesis | Superiority on infarct size          |
| PI         | Prof. Marco Valgimigli, Lugano       |
| Sponsor    | Vesalio                              |
| Registry   | NCT04969471                          |

# Inclusion criteria

- ✓ Age  $\geq 18$  years
- ✓ Chest pain  $> 20$  min with ST elevation  $\geq 1$  mm in  $\geq 2$  contiguous leads, or infero-lateral MI with ST depression  $\geq 1$  mm in  $\geq 2$  of V1–V3 and a positive terminal T wave
- ✓ TIMI thrombus grade  $\geq 3$  in the infarct-related artery (re-confirmed after wiring if TIMI 0 flow)
- ✓ Intervention started within 8 hours of symptom onset
- ✓ Informed consent before the procedure

## Key point

### **TIMI thrombus grade $\geq 3$**

A definite clot longer than half the vessel diameter. These patients are the most likely to embolise clot during PCI, so they have the most to gain from thrombectomy.

# Exclusion criteria

- ✕ Unconscious patient
- ✕ Infarct-related artery < 2.5 mm (visual estimate)
- ✕ Severe calcification or extreme tortuosity at or proximal to the culprit lesion
- ✕ Stent thrombosis as the culprit lesion
- ✕ Previous MI in the same territory
- ✕ Women of child-bearing potential
- ✕ Participation in another interventional trial



## Why these exclusions?

- Small, calcified or tortuous vessels: hard to deliver a retriever safely
- Stent thrombosis and prior MI in the same territory would confound infarct size

# Endpoints



## Primary

Infarct size by CK-MB area  
under the curve



## Secondary

Infarct size on cardiac MRI  
(% of LV mass) at day 3



## Safety

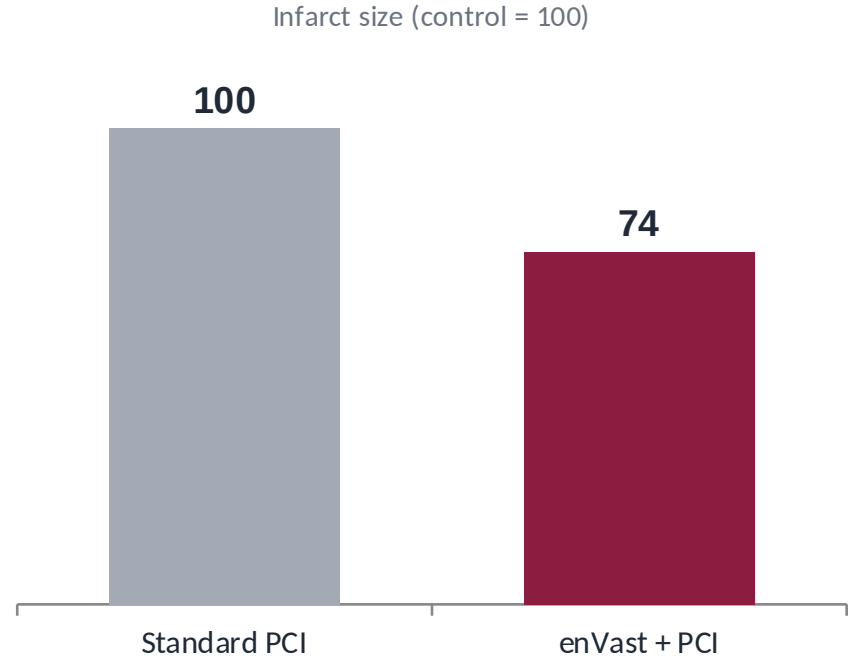
Death, stroke and MACE at  
30 days

# Primary endpoint: met

# 26%

relative reduction in infarct size (CK-MB AUC)  
with enVast versus standard PCI

**P = 0.001**



# Secondary endpoint: cardiac MRI

## CK-MB AUC (primary)

**26% smaller**

Standard PCI



enVast + PCI

*Significant,  $P = 0.001$*

## Cardiac MRI at day 3 (secondary)

**25% smaller**

Standard PCI

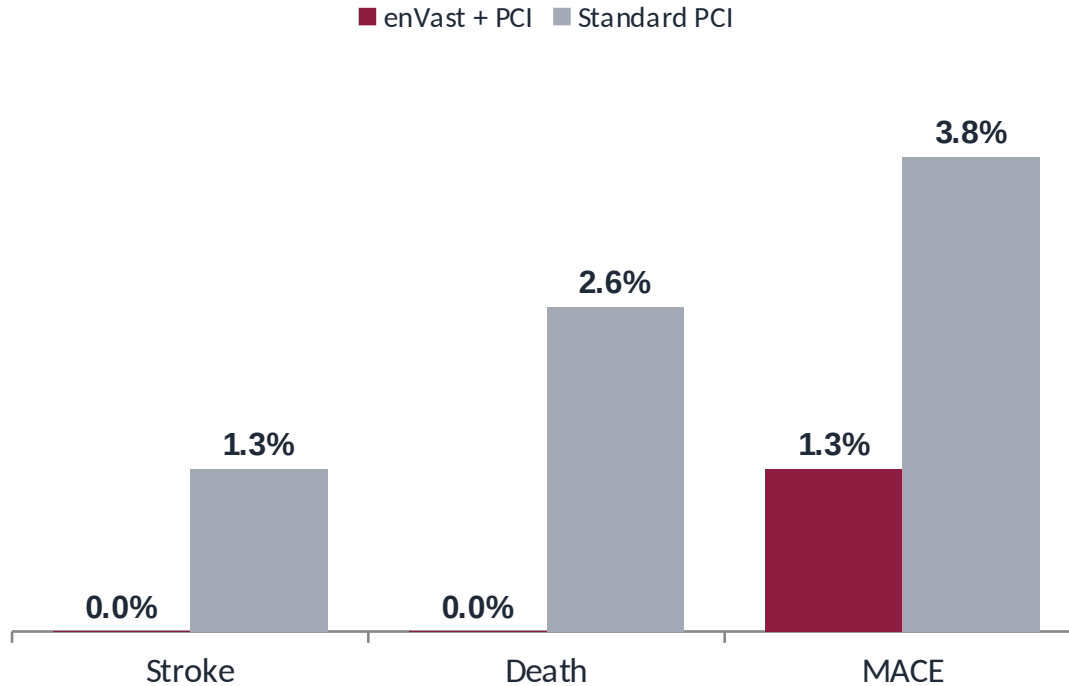


enVast + PCI

*Trend in the same direction*

*Two different measures, a blood test and imaging, point the same way.*

# Safety at 30 days



## No strokes and no deaths with enVast

- Important because aspiration in TOTAL increased stroke
- Very few events overall
- The trial was not powered for clinical outcomes

# Strengths



## **Randomised, multicentre**

11 centres, 1:1 allocation



## **Right population**

Only patients with a large thrombus



## **Consistent results**

CK-MB and cardiac MRI agree



## **Safe early course**

No stroke or death with enVast at 30 days

# Limitations

1

## **Small trial**

Only 154 patients

2

## **Surrogate endpoint**

Infarct size, not death or heart failure

3

## **Operators not blinded**

Device use cannot be hidden

4

## **MRI only a trend**

Industry-funded; needs independent confirmation

# Conclusion

*In STEMI with a large thrombus, enVast thrombectomy before PCI was superior to standard PCI in reducing infarct size, with no early safety concern.*

## Today

Promising for selected large-thrombus cases, but not yet routine. Guidelines are unchanged.

## Next

A larger trial with hard outcomes: death, heart failure and stroke.

# Take-home for our cath lab

**01**

Grade the thrombus (TIMI thrombus grade) in every STEMI

**02**

Routine manual aspiration is still not recommended

**03**

Stent-retriever thrombectomy is promising for large thrombus, but awaits outcome data

# References

1. Valgimigli M. NATURE trial. Late-Breaking Clinical Trial, ESC Congress 2026, Munich.
2. Landi A, et al. Mechanical thrombectomy in STEMI with large thrombus burden: design and rationale of the NATURE trial. Cardiovasc Revasc Med. 2026.
3. ClinicalTrials.gov NCT04969471: NATURE (enVast as an adjunct to PPCI in STEMI).
4. Fröbert O, et al. Thrombus aspiration during STEMI (TASTE). N Engl J Med. 2013.
5. Jolly SS, et al. Primary PCI with or without routine manual thrombectomy (TOTAL). N Engl J Med. 2015.
6. Byrne RA, et al. 2023 ESC Guidelines for the management of acute coronary syndromes. Eur Heart J. 2023.

# Thank you

Questions and discussion

Dr M. Waqas Saleem · PGR Cardiology · Cardiac Centre, BVH Bahawalpur